

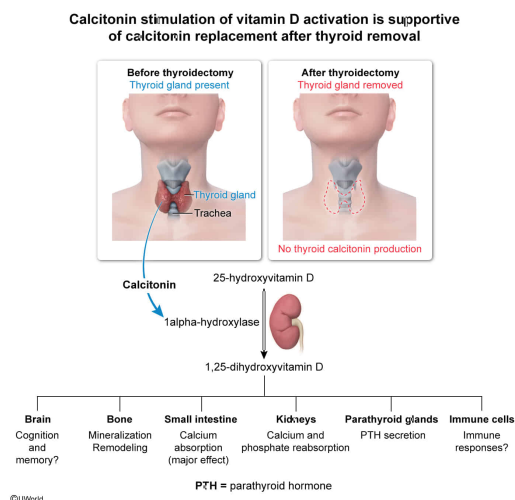
An endocrinologist hypothesizes that calcitonin should be replaced in individuals whose thyroid gland has been removed due to goiter, an enlargement of the thyroid gland. Which of the experimental findings below best supports this hypothesis?

- ☐ A. Patients who have undergone thyroidectomy do not have an increased fracture risk. [11%]
- ☐ B. Mice genetically engineered to lack calcitonin have higher bone mineral density than mice with normal calcitonin levels. [25%]
- ✓ ☒ C. Calcitonin is the primary stimulator of enzymatic vitamin D activation in the kidney. [53%]
- ☐ D. Women whose thyroids have been removed have increased calcitonin levels during pregnancy and have normal calcitonin levels in breastmilk. [9%]

Correct

53% answered correctly

Explanation:



Calcitonin is a **hormone** thought to inhibit **bone resorption** and thereby prevent excessive loss of bone density or elevation of blood calcium level. The primary site of calcitonin production is the thyroid gland.

Vitamins are essential micronutrients that can be obtained in the diet and function as **cofactors** or coenzymes in reactions in the body. **Vitamin D** is both consumed in the diet and synthesized in the skin in response to sunlight. Activating reactions in the liver and the kidney convert vitamin D from the diet or skin to **calcitriol**, the most active form of vitamin D.

Calcitriol acts as a **steroid hormone**, binding to receptors throughout the body (eg, T and B lymphocytes, mononuclear cells, brain cells). In its role as one of the mediators of calcium handling in the body, calcitriol stimulates intestinal absorption of calcium and **reabsorption** of calcium in the kidney.

The question relates an endocrinologist's hypothesis that calcitonin should be replaced in individuals whose thyroid gland has been removed and asks for the finding that would be most supportive of this hypothesis.

Calcitonin could have important effects in tissues other than bone, and such effects might suggest that calcitonin therapy is warranted after thyroidectomy. Also, if the effects of vitamin D (calcitriol) are important even in the absence of low blood calcium concentrations, then a mediator of kidney calcitriol synthesis other than **parathyroid hormone** may be required. Therefore, the finding that

calcitonin stimulates vitamin D activation by the kidney **best supports** the endocrinologist's hypothesis **that calcitonin should be replaced** in individuals whose thyroid glands have been removed.

The other options do *not* support treatment with calcitonin after thyroid removal.

(Choice A) The finding that patients who have undergone thyroidectomy do not have an increased risk of fracture suggests that calcitonin replacement is *not* required to prevent fragile bones.

(Choice B) Higher bone density in calcitonin knockout mice suggests that calcitonin is nonessential for bone health and may not be necessary to replace after thyroidectomy.

(Choice D) The finding that women whose thyroids have been removed still have increased calcitonin levels during pregnancy and have normal calcitonin levels in their breastmilk demonstrates that calcitonin can be produced *outside* of the thyroid. This finding suggests that calcitonin replacement may *not* be necessary even during special situations such as pregnancy.

Educational objective:

Calcitonin is a hormone conventionally considered to inhibit bone resorption and thereby prevent excessive blood calcium concentration and bone density loss.

Time spent:0 sec

Subject:
Biology

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System:
Endocrine and Nervous Systems